Face Mask Detection Report

1. Introduction

The Face Mask Detection System is a computer vision-based project developed to detect whether a person is wearing a face mask in real-time. During health crises like COVID-19, wearing masks became essential for safety. This project uses a webcam to monitor individuals and automatically classify them as “Mask” or “No Mask”. It helps in enforcing safety rules in public places such as offices, schools, and transport areas.

2. Abstract

This project implements a real-time face mask detection system using deep learning and computer vision techniques. A Convolutional Neural Network (CNN) model is trained on a dataset of masked and unmasked faces. The system uses OpenCV for capturing live video and Haar Cascade for detecting faces. The trained model then predicts whether the detected face is wearing a mask or not. If no mask is detected, an alert can be triggered. The system is efficient, fast, and suitable for real-time applications.

3. Tools Used

- Python
- OpenCV
- TensorFlow / Keras
- NumPy
- Haar Cascade Classifier
- Flask for deployment

4. Steps Involved in Building the Project

- Collected dataset (with mask / without mask)
- Preprocessed images (resize, normalize)
- Trained CNN model using Keras
- Detected faces using Haar Cascade
- Performed real-time mask prediction via webcam
- Displayed result (Mask / No Mask)
- Added alert for “No Mask”

5. Conclusion

- The Face Mask Detection System successfully demonstrates the use of deep learning in real-time applications. It can detect masks efficiently using a webcam and provide instant feedback. This project can be further improved by increasing dataset size, improving model accuracy, and deploying it on web or mobile platforms. It is a useful application in maintaining public safety and health standards.

Name : MD ARMAN